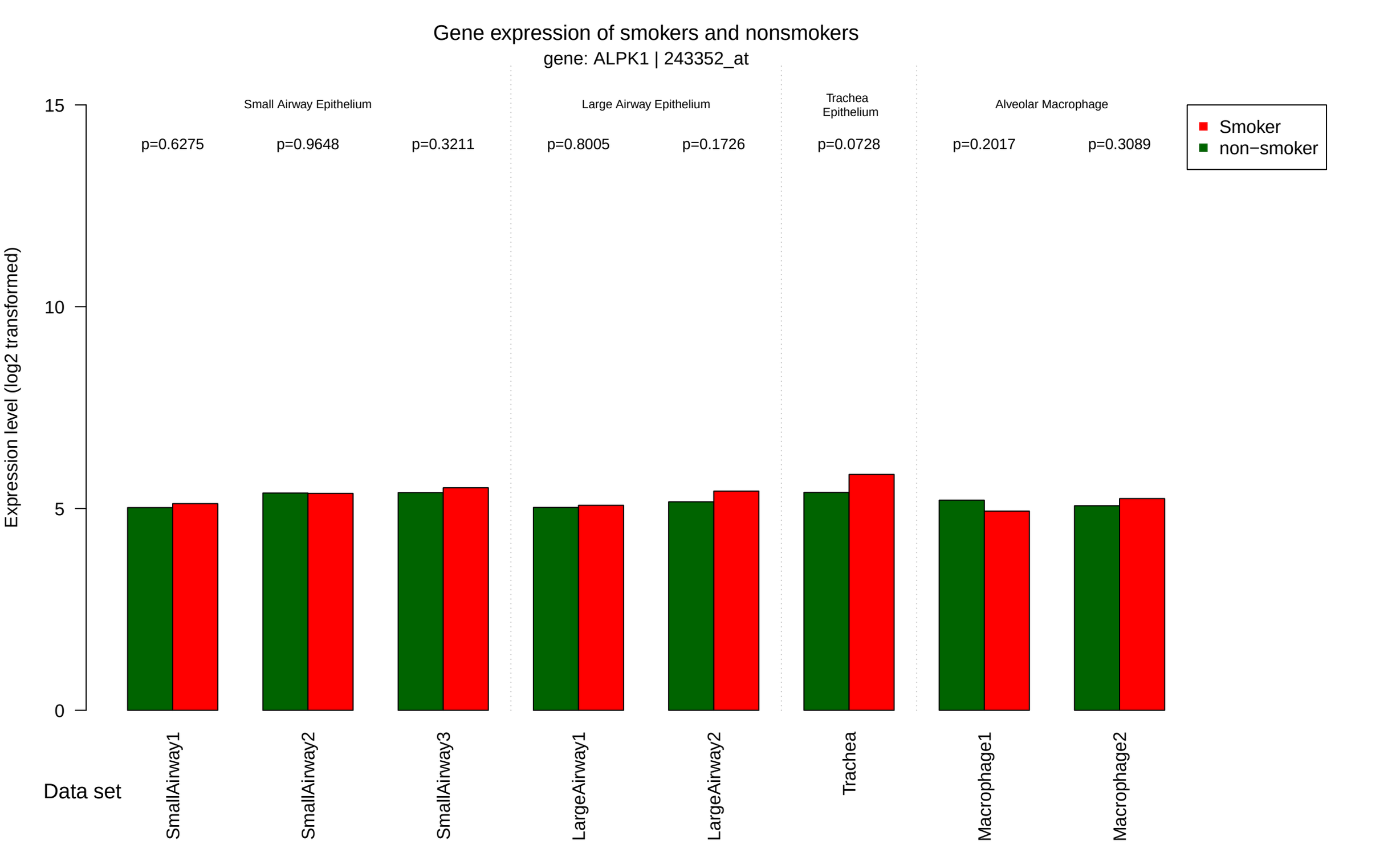
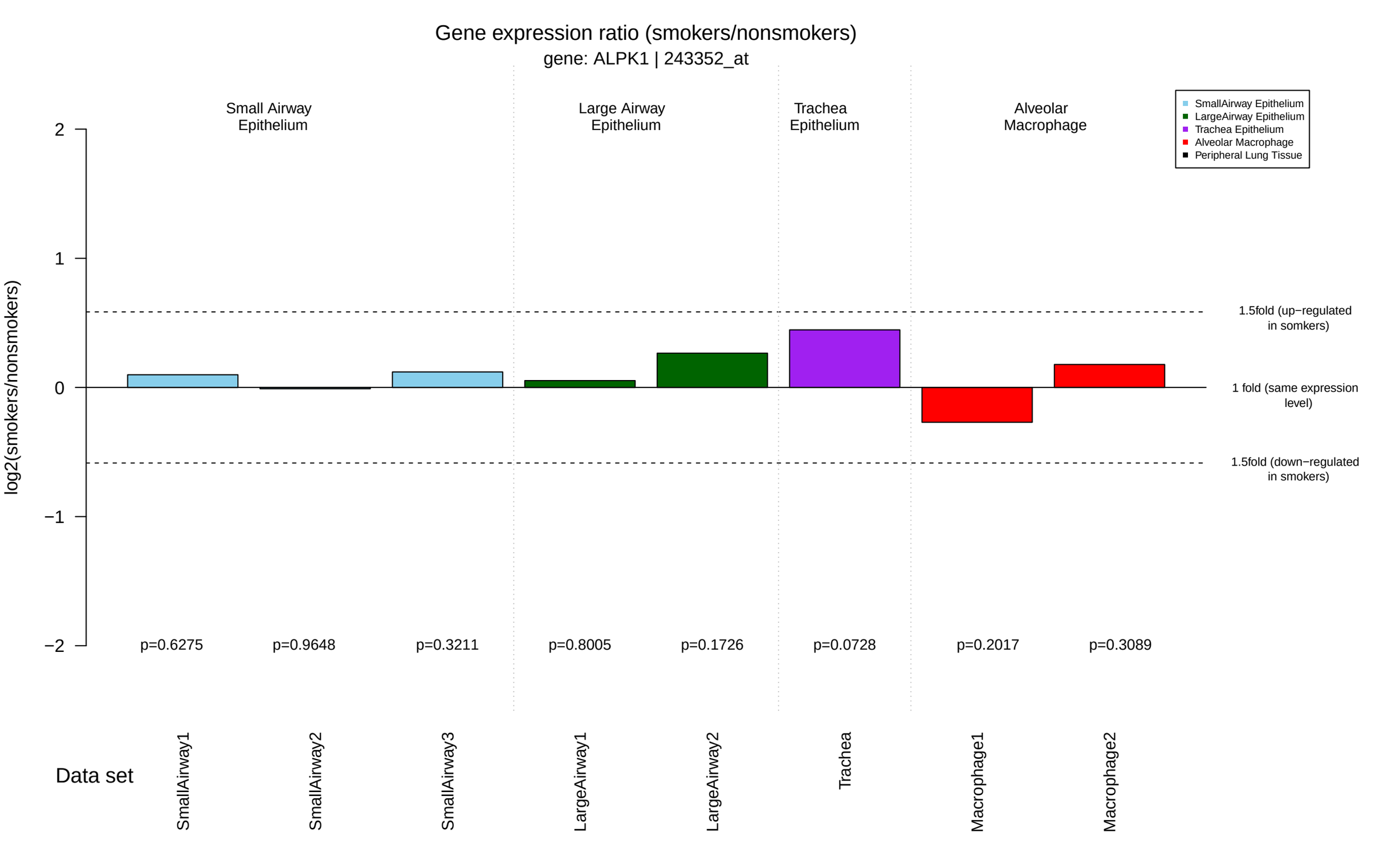
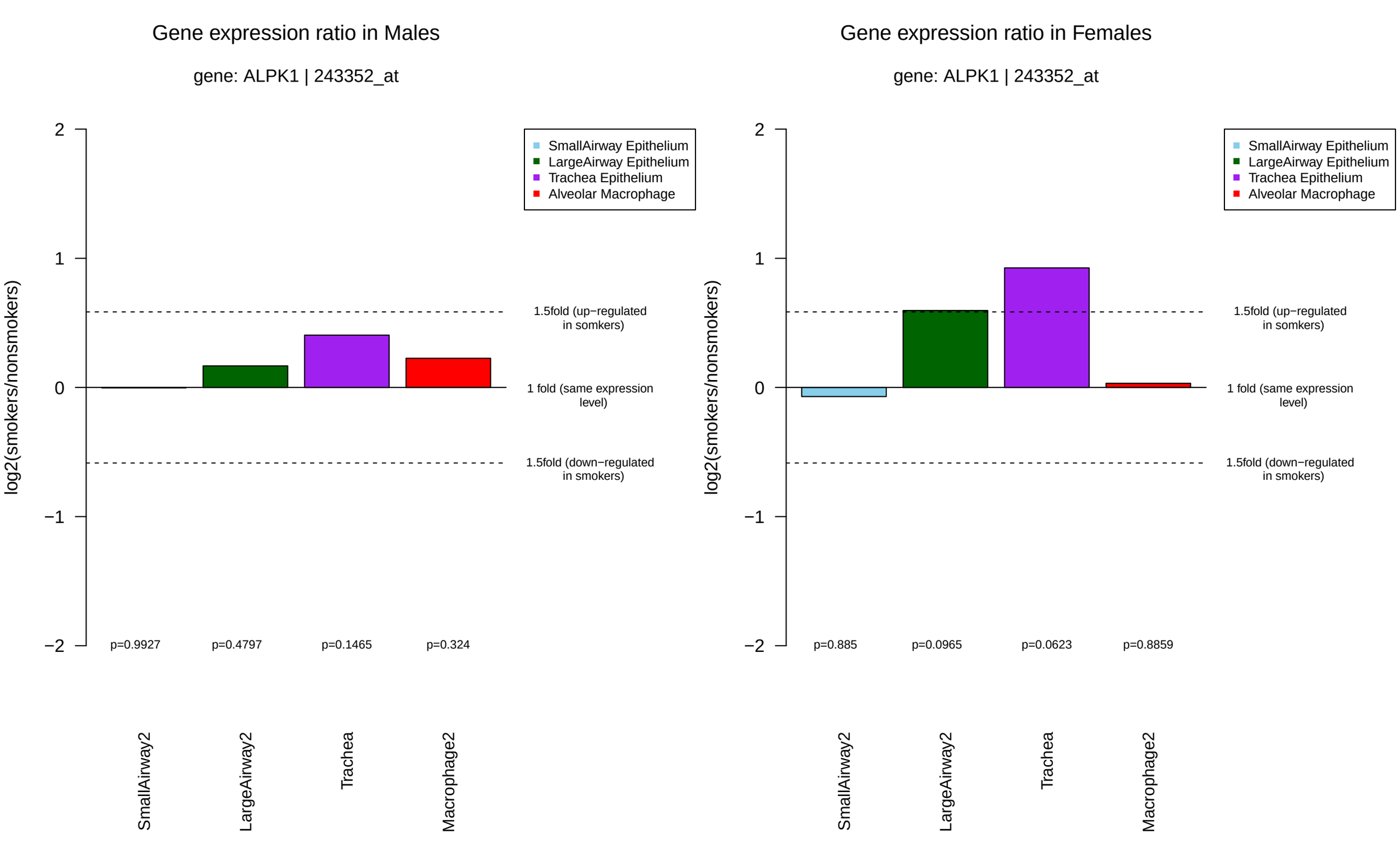
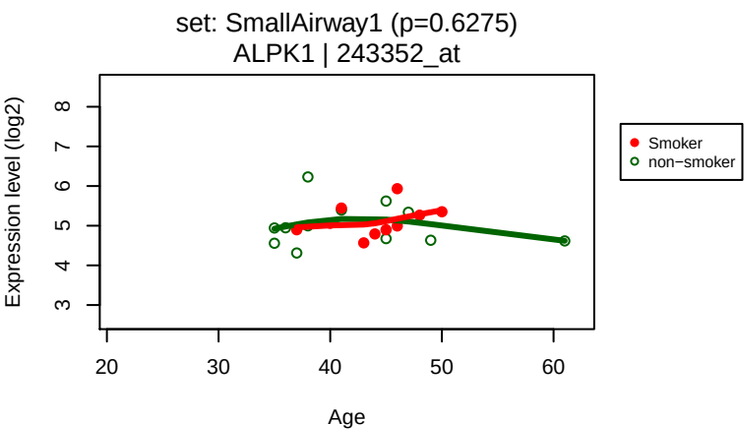
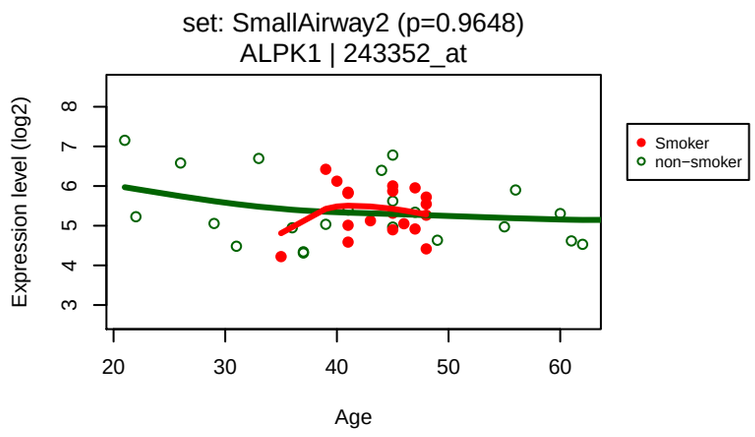
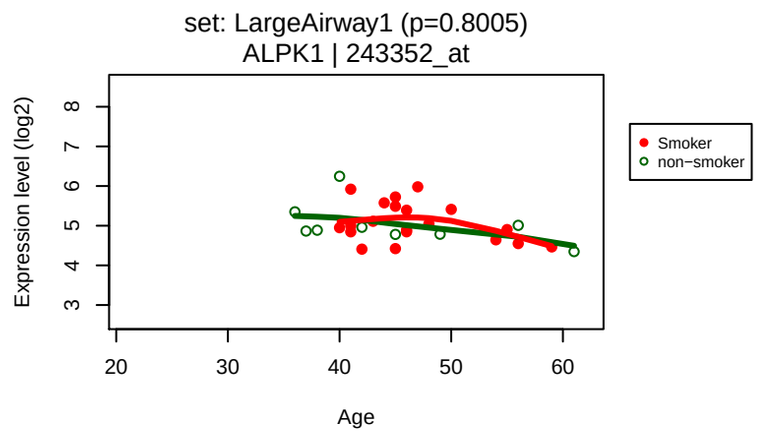

Gene expression of smokers and nonsmokers
gene: ALPK1 | 243352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6275
p=0.9648
p=0.3211
p=0.8005
p=0.1726
p=0.0728
p=0.2017
p=0.3089
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

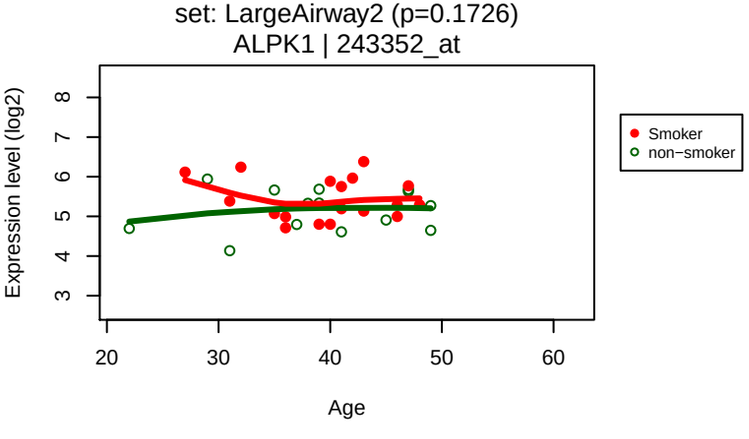

Gene expression ratio (smokers/nonsmokers)
gene: ALPK1 | 243352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6275
p=0.9648
p=0.3211
p=0.8005
p=0.1726
p=0.0728
p=0.2017
p=0.3089
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

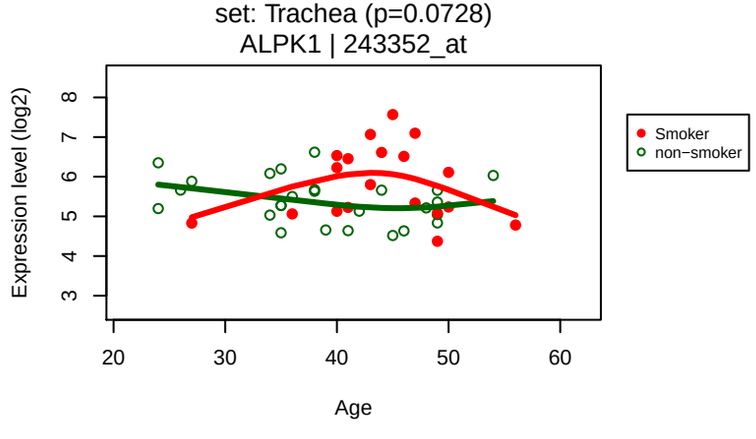

Gene expression ratio in Males
gene: ALPK1 | 243352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9927
p=0.4797
p=0.1465
p=0.324
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALPK1 | 243352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.885
p=0.0965
p=0.0623
p=0.8859
SmallAirway2
LargeAirway2
Trachea
Macrophage2

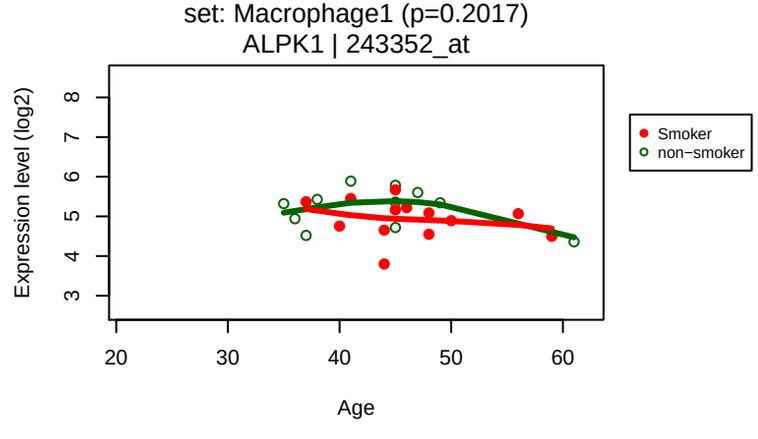

set: SmallAirway1 (p=0.6275)
ALPK1 | 243352_at
Expression level (log2)
Age
Smoker
non−smoker

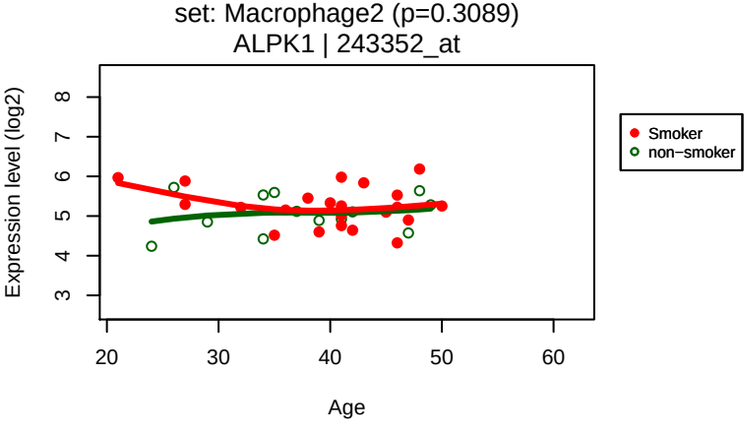

set: SmallAirway2 (p=0.9648)
ALPK1 | 243352_at
Expression level (log2)
Age
Smoker
non−smoker

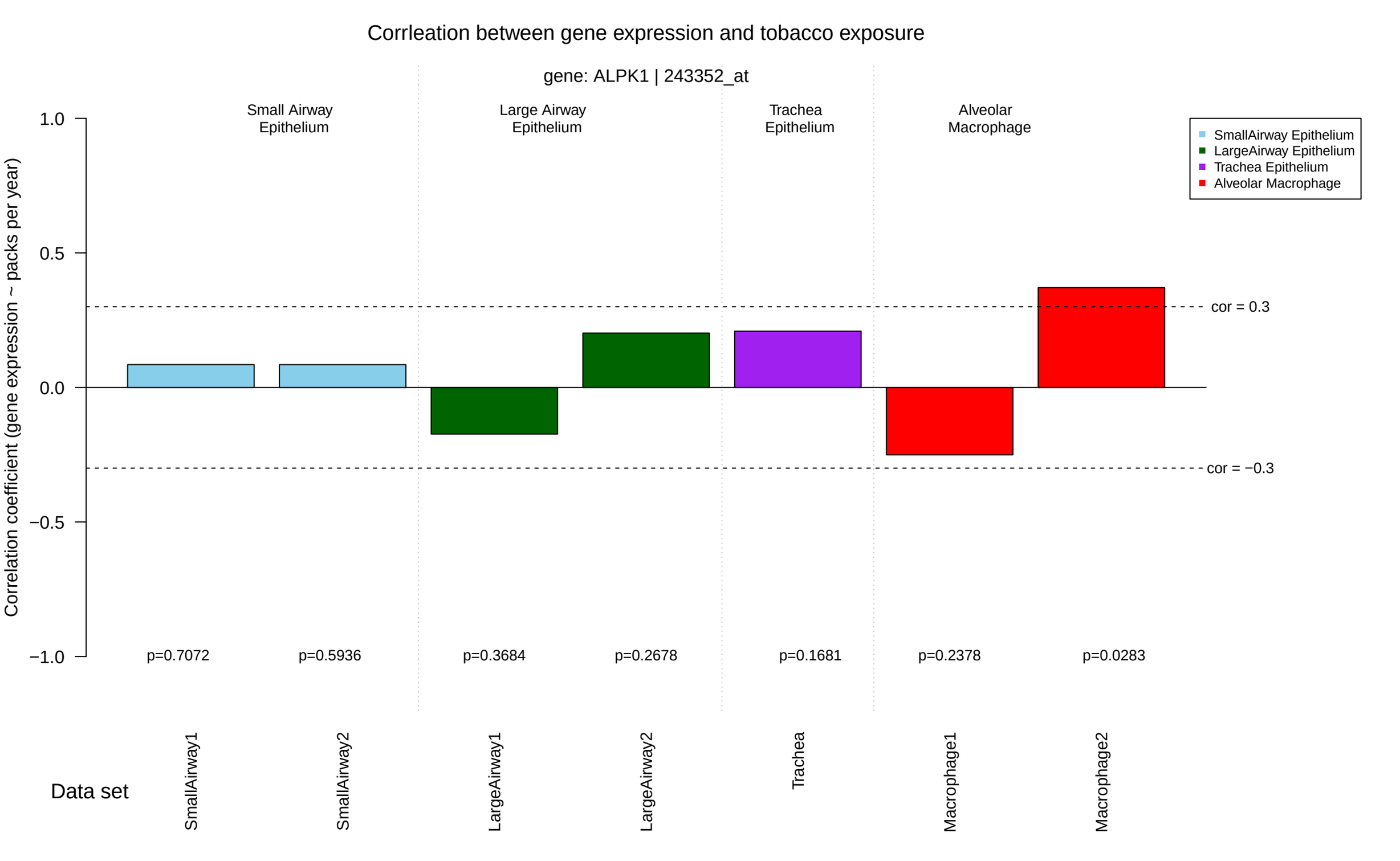

set: LargeAirway1 (p=0.8005)
ALPK1 | 243352_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1726)
ALPK1 | 243352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0728)
ALPK1 | 243352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2017)
ALPK1 | 243352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3089)
ALPK1 | 243352_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALPK1 | 243352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7072
p=0.5936
p=0.3684
p=0.2678
p=0.1681
p=0.2378
p=0.0283
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2